

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 1 Practical No: 2

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Subject Name : Deep Learning

Assignment 2: Design and implement a Multilayer Perceptron (MLP) for classification of the Iris or Wine dataset, and evaluate its performance using accuracy and a confusion matrix.

Algorithm :

1. Import the required libraries such as NumPy, Pandas, Matplotlib, Seaborn, TensorFlow/Keras, and Scikit-learn.
2. Load the Iris dataset using Scikit-learn.
3. Convert the dataset into a Pandas DataFrame for easy analysis.
4. Perform Exploratory Data Analysis (EDA) by checking:
 1. Dataset shape
 2. First few records
 3. Data types
 4. Statistical summary
5. Separate the features (X) and target labels (y).
6. Split the dataset into training and testing sets using an 80:20 ratio.
7. Normalize the feature values using StandardScaler.
8. Build a Sequential Multilayer Perceptron (MLP) model consisting of :
 1. Input layer
 2. Two hidden Dense layers with ReLU activation
 3. Output Dense layer with Softmax activation
9. Compile the model using:
 1. Adam Optimizer
 2. Sparse Categorical Crossentropy Loss
 3. Accuracy Metric
10. Train the model using the training dataset for a fixed number of epochs.
11. Evaluate the trained model on the testing dataset.

12. Predict the class labels of the testing data.
13. Calculate the classification accuracy.
14. Generate the Confusion Matrix.
15. Display the Confusion Matrix using a heatmap.
16. Analyze the obtained accuracy and confusion matrix.

Code :

Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

Load Dataset

```
iris = load_iris()
X = iris.data
y = iris.target
```

Create DataFrame

```
df = pd.DataFrame(X, columns=iris.feature_names)
df["Species"] = y
print(df.head())
```

Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

Feature Scaling

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Build MLP Model

```
model = Sequential()
model.add(Dense(16, activation='relu', input_shape=(4,)))
```

```

model.add(Dense(8, activation='relu'))
model.add(Dense(3, activation='softmax'))

# Compile Model
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

# Train Model
history = model.fit(
    X_train,
    y_train,
    epochs=50,
    batch_size=8,
    verbose=1
)

# Evaluate Model
loss, accuracy = model.evaluate(X_test, y_test)

# Prediction
y_pred = model.predict(X_test)
y_pred = np.argmax(y_pred, axis=1)

# Accuracy Score
acc = accuracy_score(y_test, y_pred)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

# Plot Confusion Matrix
plt.figure(figsize=(6,5))
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

```

Output

Sample dataset

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	Target
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

Shape: (150, 5)

Model training

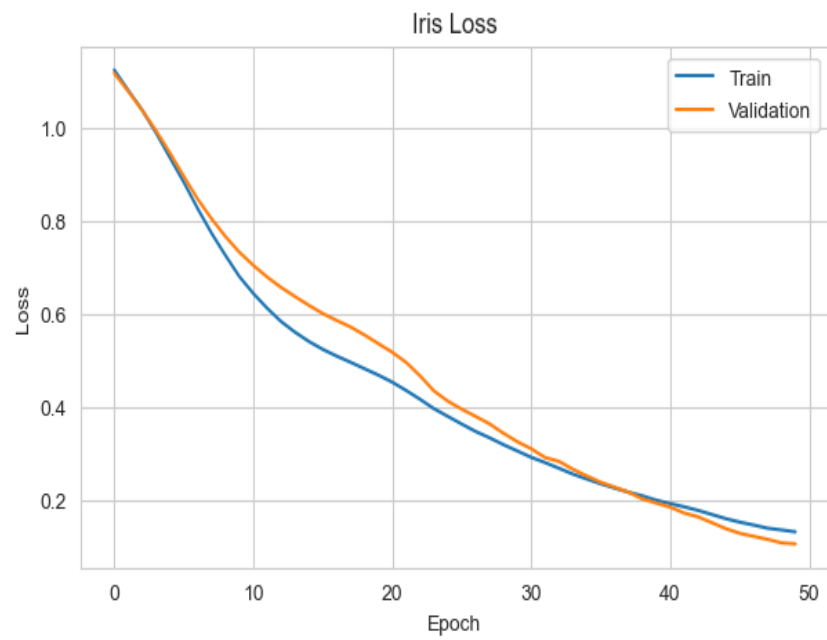
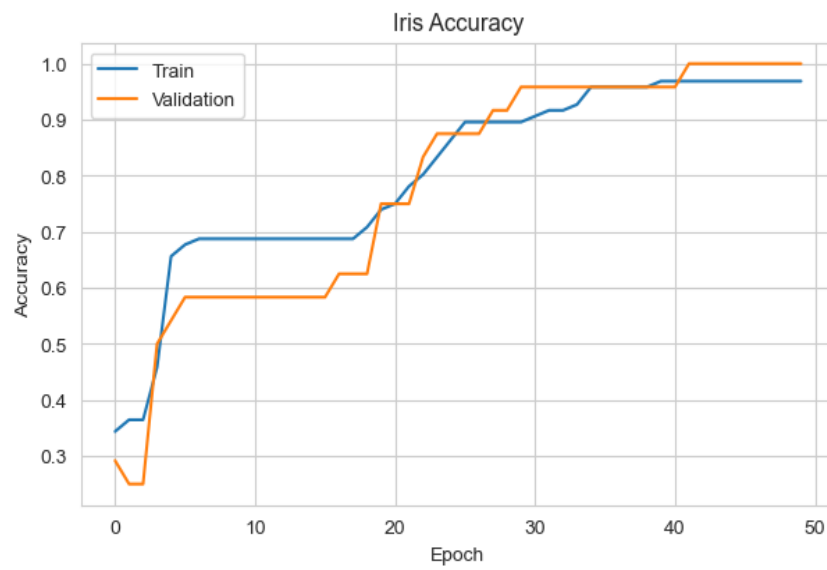
12/12	0s 8ms/step	- accuracy: 0.9688 - loss: 0.1541 - val_accuracy: 1.0000 - val_loss: 0.1301
Epoch 47/50		
12/12	0s 16ms/step	- accuracy: 0.9688 - loss: 0.1479 - val_accuracy: 1.0000 - val_loss: 0.1237
Epoch 48/50		
12/12	0s 12ms/step	- accuracy: 0.9688 - loss: 0.1411 - val_accuracy: 1.0000 - val_loss: 0.1172
Epoch 49/50		
12/12	0s 30ms/step	- accuracy: 0.9688 - loss: 0.1374 - val_accuracy: 1.0000 - val_loss: 0.1097
Epoch 50/50		
12/12	0s 15ms/step	- accuracy: 0.9688 - loss: 0.1334 - val_accuracy: 1.0000 - val_loss: 0.1075

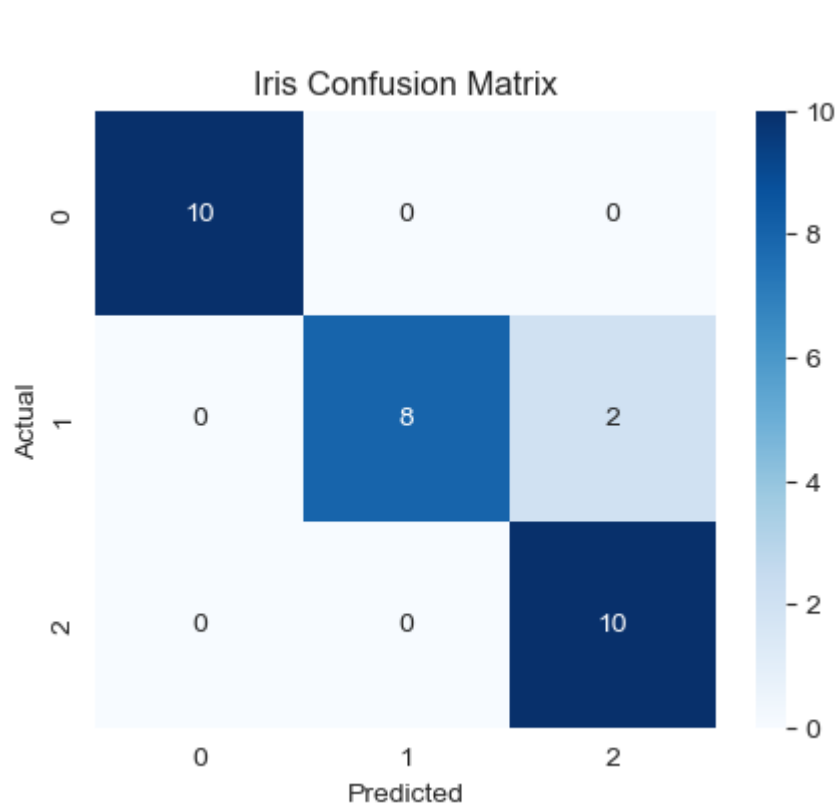
...

Model: "sequential"

...

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 16)	80
dense_1 (Dense)	(None, 8)	136
dense_2 (Dense)	(None, 3)	27





Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	0.80	0.89	10
2	0.83	1.00	0.91	10
accuracy			0.93	30
macro avg	0.94	0.93	0.93	30
weighted avg	0.94	0.93	0.93	30